

Suggested Questions are 2, 3, 4 and 5

Qu. 1. a) By considering the Taylor expansion for $1/(1-w)$ (about 0), where $|w| < 1$, show that

$$1/(1+t^2) = (1 - t^2 + t^4 - \dots).$$

Under what conditions is this expansion valid?

- b) By integrating this series term by term, find an expansion for $\arctan(u)$
 c) Find the Taylor expansion (up to and including cubic terms) of the function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ about the point $(1, 1, -1)$ where $f(x, y, z) = \arctan(x + y + 2z)$.

Qu. 2. a) Given that the Taylor series for $\sin x$ about 0 is

$$\sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

find the Taylor series for $\sin(x+y)$ about $(0, 0)$.

- b) By using the fact that, for $|w| < 1$,

$$\frac{1}{1-w} = \sum_{n \geq 0} w^n,$$

find the Taylor series of $\frac{1}{1 + \sin(x+y)}$ up to and including cubic terms.

Qu. 3. Let P be the tangent plane to the surface $xyz = a^3$ at the point (r, s, t) . Find the Cartesian equation of P .

Let T be the tetrahedron formed by the coordinate planes, $x = 0$, $y = 0$, $z = 0$, and the tangent plane P . Show that the volume of T is independent of the point (r, s, t) . Recall that the volume of a tetrahedron is $\frac{1}{3}(\text{area of base} \times \text{height})$.

Qu. 4. Find critical points of the following functions:

- a) $z = x^3 + y^3 - 3xy$ (determine the type of critical points)
 b) $z = xy\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$, $a > 0$, $b > 0$.

Qu. 5. Find the critical points of the function $f(x, y) = \sin x \sin y \sin(x+y)$ in an open domain $D = \{x, y : 0 < x < \pi, 0 < y < \pi\}$ and determine their type.